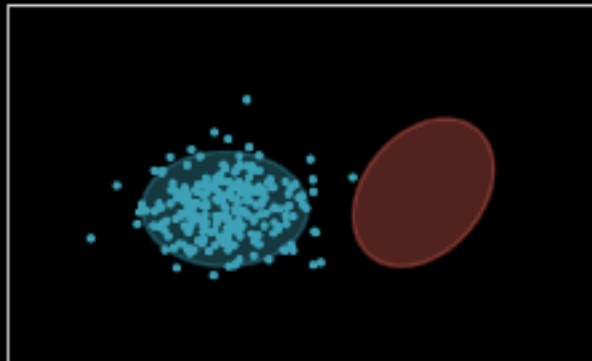


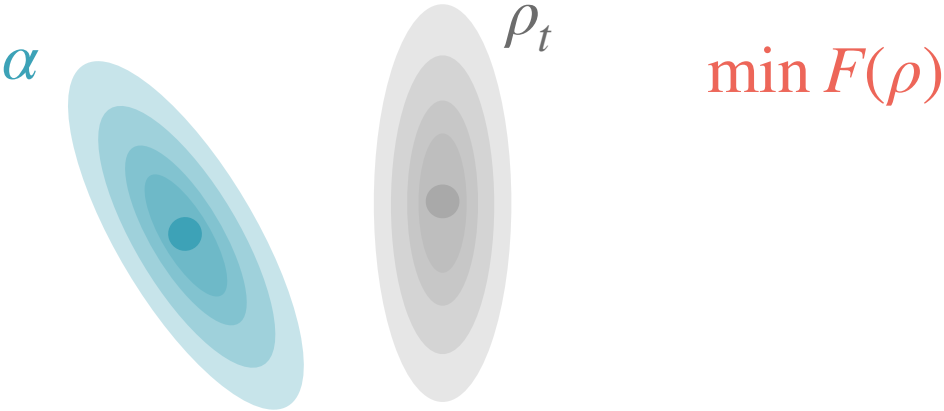
Schroedinger bridge between gaussians



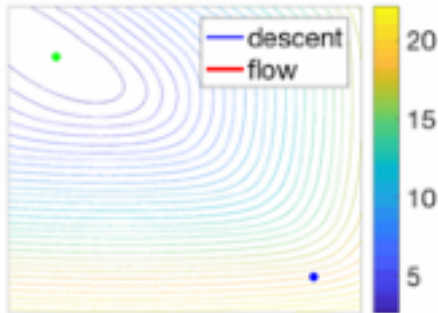
solve problem gradually
functional $F(\rho)$







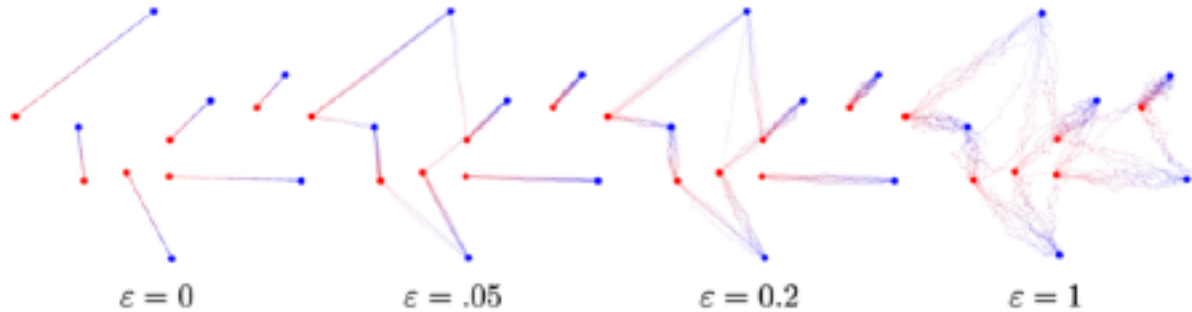
$k = 0$



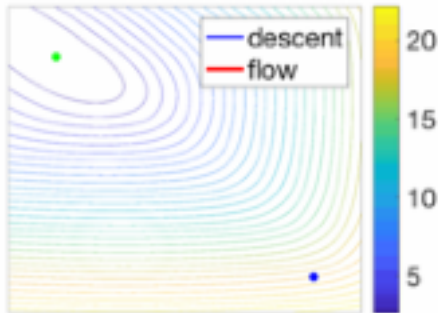


also stochastic/diffuse paths

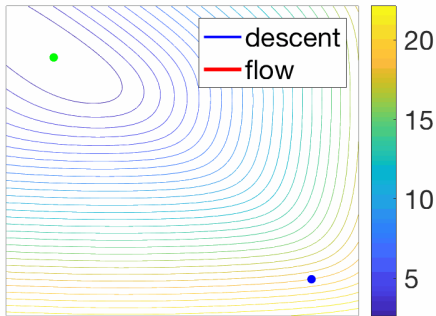
**most likely
stochastic evolution**



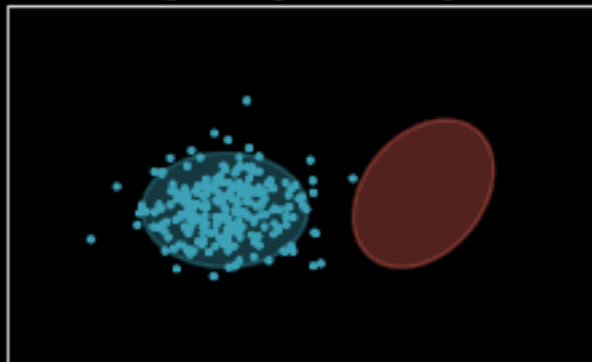
$k = 0$



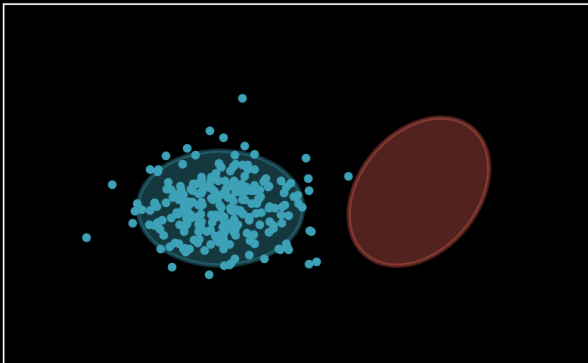
$k = 0$



Schroedinger bridge between gaussians



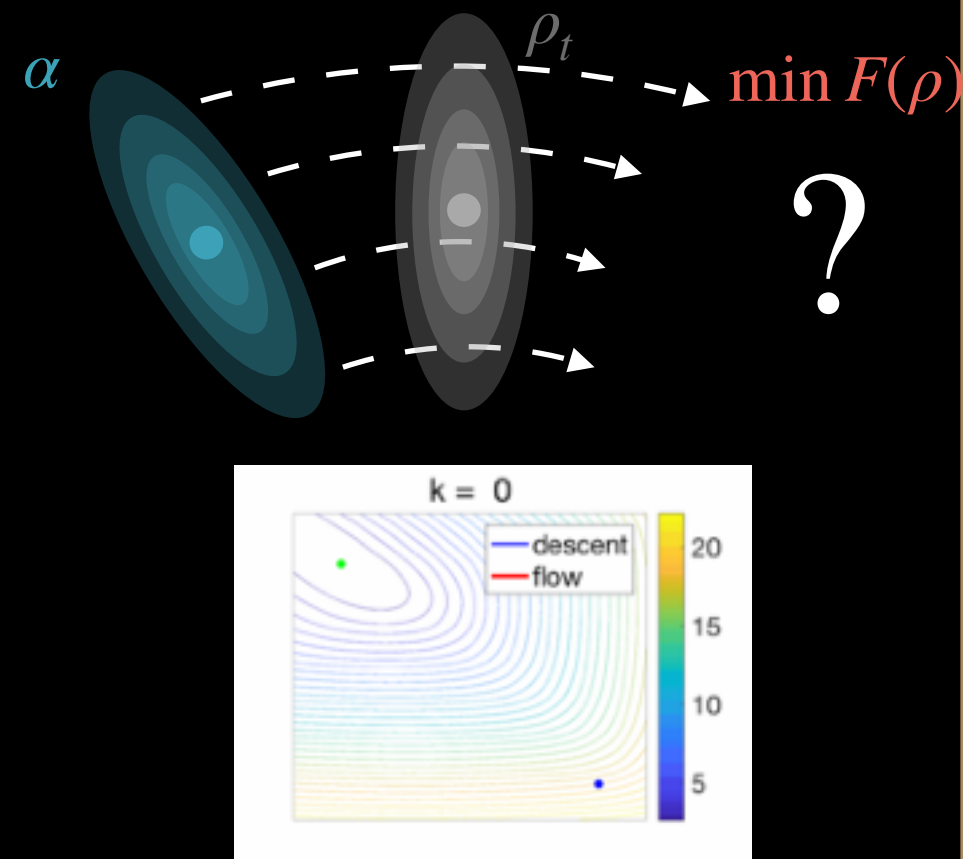
Schroedinger bridge between gaussians



Last Time

Gradient Flows

- Motivation: often we can't compute entire dynamic OT path, or don't know β
- Instead: **solve problem gradually** by minimizing **functional** $F(\rho)$
- Gradient Flow** $\sim$ limit of step size $\rightarrow 0$ of grad. descent (Euler), defines continuous curve starting at x_0 that follows $-\nabla f$
- Easy in Euclidean space! Two types of discretization: implicit, explicit.

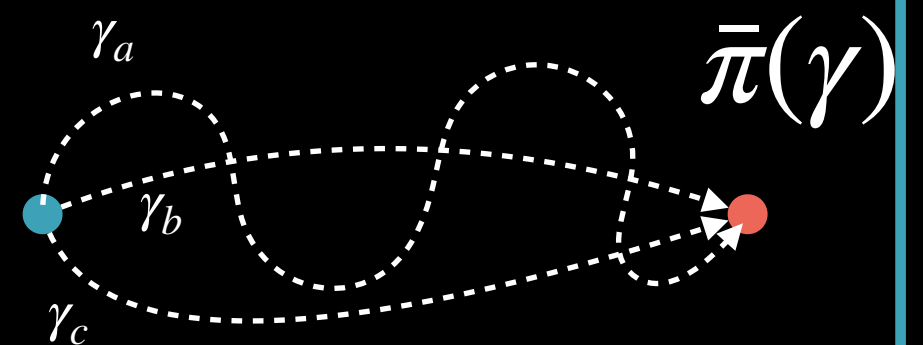


Dynamic Formulation on Path Space

- Dynamic OT over the space of paths:

$$W_2^2(\rho_0, \rho_1) = \min_{\bar{\pi} \in \bar{\Pi}} \int_{\bar{X}} L(\gamma)^2 d\bar{\pi}(\gamma)$$

where $L(\gamma) = \int_0^1 |\dot{\gamma}(s)|^2 ds = \text{path length}$



- Optimal $\bar{\pi}^*$ equivalent to Kantorovich solution (mass only on geodesics)
- Snapshots $\rho_t \triangleq (P_t)_\# \bar{\pi}^*$ recover Benamou-Brenier solution

JKO Scheme

$\sim$ Implicit Euler time-stepping in probability space:

$$\rho_{t+1} = \arg \min_{\rho} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho_t) \right\}$$

↓ Limit as $\tau \rightarrow 0$

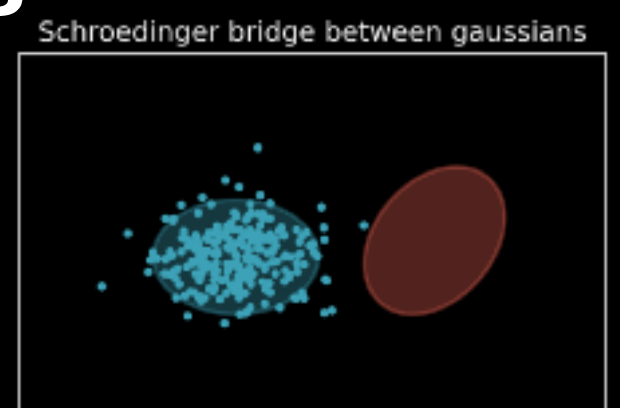
Continuous Gradient Flow (PDE)

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right)$$

Recovers many well-known PDEs (heat, F-P) for choices of $\mathcal{F}$

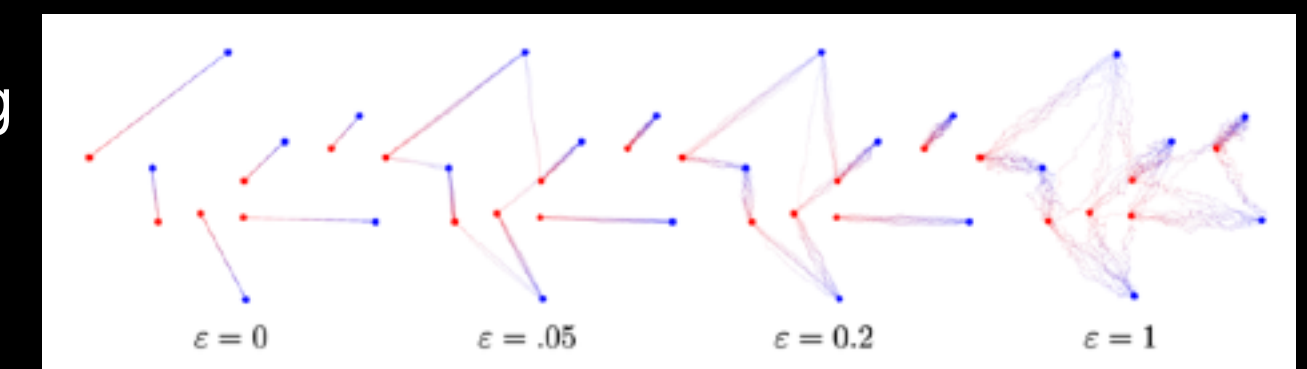
Schrödinger Bridges

- Intuition: Consider not only straight paths (geodesics) but **also stochastic/diffuse paths**.
- Regularize $\bar{\pi}$ by enforcing closeness to a reference diffuse path measure (brownian motion)



- Schrödinger Bridge problem: $\min_{\bar{\pi} \in \mathcal{U}(\rho_0, \rho_1)} \text{KL}(\bar{\pi} \parallel \bar{\mathcal{K}})$

- Interpretation: **most likely stochastic evolution** matching marginals.
- Equivalent to Entropic OT: Can be solved with Sinkhorn



Today